

Math 104 – Real Analysis

Homework 2 — my solutions

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Homework 2

Deadline: 11:59 pm September 13, 2026

Problem (1). Let S be a nonempty bounded subset of $\mathbb{R}$, and define the diameter of S by

$$\text{diam}(S) = \sup\{|s - t| : s, t \in S\}.$$

Prove that $\text{diam}(S) = \sup S - \inf S$.

Problem (2). Prove that for every real number $a > 0$ there is a unique natural number n such that

$$n - 1 \leq a < n.$$

(Your proof will need both the Archimedean Property and the Well-Ordering Principle for $\mathbb{N}$, together with an argument for uniqueness.)

Problem (3). Let

$$S = \{x \in \mathbb{R} : x^4 - 5x^2 + 4 < 0\}.$$

(a) Write S in interval notation.

(b) Find $\inf S$ and $\sup S$, and state clearly whether each is attained.

Problem (4). For each sequence below, determine whether it converges and, if so, find its limit. (You may use the limit theorems; no ε - N proof is required here, but you must justify divergence when it occurs, e.g. by exhibiting two subsequences with different limits.)

$$(a) a_n = \frac{2n^3 + n}{n^3 - 5}, \quad (b) b_n = (-1)^n \frac{n}{n + 1}.$$

Problem (5). Give rigorous ε - N proofs of the following limit.

$$\lim_{n \rightarrow \infty} \frac{3n^2 - 1}{2n^2 + n + 5} = \frac{3}{2}.$$

Solutions

Problem 1

Solution.



Problem 2

Solution.



Problem 3

Solution.



Problem 4

Solution.



Problem 5

Solution.

